

The presence of so many binding constraints on bank balance sheets has a number of important implications for the fixed income markets, two of which are discussed in Chapter 18. First, relative value analysts must incorporate these shadow costs into their analysis as and when these constraints are binding on the balance sheets of the firms for which they work. Second, analysts must incorporate the fact that traders at other firms will often face the same sorts of shadow costs and that these generally won't be observable in the market. In the old days, it wasn't difficult to produce a reasonable estimate of the marginal costs that other market participants faced when analyzing certain trades. These days, we typically have very little understanding of the shadow costs faced by traders at the largest banks.

As a result, it has become more difficult for relative value analysts to produce useful forecasts about the eventual richening or cheapening of various instruments. Given that the problem of unobservable input variables exists independently of the sophistication of the model, the analytic ideal to explain and predict all pricing relationships via no-arbitrage models needs to be reviewed. While these relationships continue to provide important insights, their influence on market prices on a given day depends on the unobservable shadow cost of arbitrage capital of the marginal market participant.

Chapter 18 highlights one of the more notable instances of this phenomenon: the repo spike of September 2019. As Jamie Dimon later stated publicly, JP Morgan wasn't in a position to satisfy the demand for cash against collateral that day, due to binding regulatory constraints on its balance sheet. He also noted that this was a change for the bank and that the bank had provided additional funds in similar situations prior to these additional regulations. We presume other banks faced similar constraints on their balance sheets at the same time. Repo rates are fundamental to a large number of relative value trades, and all of these are affected in some way when repo rates spike as high as they did that day. Relative value analysts have no choice but to grapple with these shadow costs and their effects on pricing in the fixed income markets, despite the fact that it can be a difficult task.

COMPUTATIONAL ACCESS TO COMPLEX MODELS

Beside these policy-related changes, over the last decade, even more computational capabilities have become available to the average market participant. Much of these have been directed at machine learning and artificial intelligence, including efforts to find attractive opportunities in fixed income markets. As far as we are aware, the widespread benefits of these approaches are still uncertain. We have heard reports of some firms generating attractive returns with these approaches and have been pleased with a few of our own

ventures in this area. But, in general, we're still unclear whether the benefits of mass adoption of these approaches in the fixed income markets will produce the results for which so many analysts are working.

On the other hand, we have also used the enhanced computational capabilities available these days to make use of statistical models that may have been too slow or cumbersome for the average analyst in the past. One of these is the multivariate Ornstein–Uhlenbeck (MVOU) process – a generalization of the univariate Ornstein–Uhlenbeck process discussed in the first edition of this book and appearing again in Chapter 2 of this edition. The MVOU process has some distinct benefits when applied to financial markets, but it also comes with some distinct costs. In particular, it involves numerical optimizations over a larger number of parameters, which can take some time. But as computational speeds improve, the times required to perform these optimizations decrease, particularly when making use of the shortcuts we discuss in Chapter 4 for initializing the optimization algorithm with a guess that is likely to be in the vicinity of the optimized parameter estimates.

With change comes opportunity. But to benefit from opportunity, we need to respond effectively to change. This second edition of *Fixed Income Relative Value Analysis* represents our response to the changes that have occurred in the markets in recent years. We hope that readers are able to use the ideas in this edition to profit from the opportunities afforded by these changes.

Relative Value

THE CONCEPT OF RELATIVE VALUE

Relative value is a quantitative analytical approach toward financial markets based on two fundamental notions of modern financial economics.

Proposition 1: If two securities have identical payoffs in every future state of the world, then they should have identical prices today.

Violation of this principle would result in the existence of an arbitrage opportunity, which is inconsistent with equilibrium in financial markets.

This proposition seems relatively straightforward now, but this wasn't always the case. In fact, Kenneth Arrow and Gérard Debreu won Nobel prizes in economics in 1972 and 1983 in part for their work establishing this result. And Myron Scholes and Robert Merton later won Nobel prizes in economics in 1997 for applying this proposition to the valuation of options. In particular, along with Fischer Black, they identified a self-financing portfolio that could dynamically replicate the payoff of an option, and they were able to determine the value of this underlying option by valuing this replicating portfolio.

Most of the financial models discussed in this book are based on the application of this proposition in various contexts.

Proposition 2: If two securities present investors with identical risks, they should offer identical expected returns.

This result may appear intuitive, but it's somewhat more difficult to establish than the first result. Of particular interest for our purposes is that the result can be established via the *Arbitrage Pricing Theory*, which assumes the existence of unobservable, linear factors that drive returns.

In this case, it's possible to combine securities into portfolios that expose investors to any one of the risk factors without involving exposure to any of the other risk factors. In the limit, as the number of securities in the portfolio increases, the security-specific risks can be diversified away. And in this case, any security-specific risk that offered a non-zero expected return would present investors with an arbitrage opportunity, at least in the limit, as the remaining risk factors could be immunized by creating an appropriate portfolio of tradable securities.

For our purposes, this is a powerful result, as it allows us to analyze historical data for the existence of linear factors and to construct portfolios that expose us either to these specific factors or to security-specific risks, at our discretion. In fact, *principal component analysis* (PCA) can be applied directly in this framework, and we'll rely heavily on PCA as one of the two main statistical models we discuss in this book.

THE SOURCES OF RELATIVE VALUE OPPORTUNITIES

From these two propositions, it's clear that the absence of arbitrage is the assumption that drives many of the models we use as relative value analysts. This should come as no surprise, since one of the main roles of a relative value analyst is to search for arbitrage opportunities.

But for some people, this state of affairs presents a bit of a paradox. *If our modeling assumptions are correct about the absence of free lunches, why do analysts and traders search so hard for them?*

This apparent paradox can be resolved with two observations. The first is the recognition that arbitrage opportunities are rare precisely because hard-working analysts invest considerable effort trying to find them. If these opportunities could never be found, or if they never generated any profits for those who found them, analysts would stop searching for them. But in this case, opportunities would reappear, and analysts would renew their search for them as reports of their existence circulated.

The second observation that helps resolve this paradox is that even seemingly riskless arbitrage opportunities carry some risk when pursued in practice. For example, one of the simpler arbitrages in fixed income markets is the relation between bond prices, repo rates, and bond futures prices. If a bond futures contract is too rich, a trader can sell the futures contract, buy the bond, and borrow the purchase price of the bond in the repo market, with the bond being used as collateral for the loan. At the expiration of the contract, the bond will be returned to the trader by his repo counterparty, and the trader can deliver the bond into the futures contract. In theory, this would allow the

trader to make a riskless arbitrage profit. But, in practice, there are risks to this strategy.

For example, the repo counterparty may fail to deliver the bonds to the trader promptly at the end of the repo transaction, in which case the trader may have difficulty delivering the bonds into the futures contract. Failure to deliver carries significant penalties in some cases, and the risk of incurring these penalties needs to be incorporated into the evaluation of this seemingly riskless arbitrage opportunity.

These perspectives help us reconcile the existence of arbitrage opportunities in practice with the theoretical assumptions behind the valuation models we use. But they don't explain the sources of these arbitrage or relative value opportunities, and we'll discuss a few of the more important sources here.

Demand for Immediacy

In many cases, relative value opportunities will appear when some trader experiences an unusually urgent need to transact, particularly in large size. Such a trader will transact his initial business at a price that reflects typical liquidity in the market. But if the trader then needs to transact additional trades in the same security, he may have to entice other market participants to provide the necessary liquidity by agreeing to transact at a more attractive price. For example, he may have to agree to sell at a lower price or to buy at a higher price than would be typical for that security. In so doing, this trader is signaling a demand for immediacy in trading, and he's offering a premium to other traders who can satisfy this demand.

The relative value trader searches for opportunities in which he can be paid attractive premiums for satisfying these demands for immediacy. He uses his capital to satisfy these demands, warehousing the securities until he can liquidate them at more typical prices, being careful to hedge the risks of the transactions in a cost-effective and prudent manner.

Because these markets are so competitive, the premiums paid for immediacy are often small relative to the sizes of the positions. As a result, the typical relative value fund will be run with leverage that is higher than the leverage of, say, a global macro fund. Consequently, it's important to pay attention to small details and to hedge risks carefully.

Misspecified Models

It sometimes happens that market participants overlook relevant issues when modeling security prices, and the use of misspecified models can result in attractive relative value opportunities for those who spot these errors early.

For example, until the mid-1990s, most analysts failed to incorporate the convexity bias when assessing the relative valuations of Eurodollar futures contracts and forward rate agreements. As market participants came to realize the importance of this adjustment, the relative valuations of these two instruments changed over time, resulting in attractive profits for those who had identified this issue relatively early.

In recent years, as credit concerns have increased for many governments, it has become increasingly important to reflect sovereign credit risk as an explicit factor in swap spread valuation models, and we discuss this issue in considerable detail in this book.

Regulatory Arbitrage

The fixed income markets are populated by market participants of many types across many different regulatory jurisdictions, and the regulatory differences between them can produce relative value opportunities for some.

For example, when thinking about the relative valuations of unsecured short-term loans and loans secured by government bonds in the repo market, traders at European banks will consider the fact that the unsecured loan will attract a greater regulatory charge under the Basel accords. On the other hand, traders working for money market funds in the US won't be subject to the Basel accords and are likely to focus instead on the relative credit risks of the two short-term deposits. The difference in regulatory treatment may result in relative valuations that leave the European bank indifferent between the two alternatives but that present a relative value opportunity for the US money market fund.

THE INSIGHTS FROM RELATIVE VALUE ANALYSIS

In some sense, relative value analysis can be defined as the process of gaining insights into the relationships between different market instruments and the external forces driving their pricing. These insights facilitate arbitrage trading, but they also allow us more generally to develop an understanding of the market mechanisms that drive valuations and of the ways seemingly different markets are interconnected.

As a consequence, relative value analysis, which originated in arbitrage trading, has a much broader scope of applications. It can reveal the origins of certain market relations, the reasons a security is priced a certain way, and the relative value of this pricing in relation to the prices of other securities. And in the event that a security is found to be misvalued, relative value analysis suggests ways in which the mispricing can be exploited through specific

trading positions. In brief, relative value analysis is a prism through which we view the machinery driving market pricing amidst a multitude of changing market prices.

As an example, consider the divergence of swap spreads for German Bunds and US Treasuries during the financial crisis, which might appear inextricable without considering the effects of cross-currency basis swaps (CCBS), intra-currency basis swaps (ICBS), and credit default swaps (CDS).

In this case, CCBS spreads widened as a result of the difficulties that European banks experienced in raising USD liabilities against their USD assets. On the other hand, arbitrage between Bunds, swapped into USD, and Treasuries prevented an excessive cheapening of Bunds versus USD LIBOR. As a consequence, Bunds richened significantly against EURIBOR (see Chapter 16 for more details).

However, given the relationship between European banks and sovereigns, the difficulties of European banks were also reflected in a widening of European sovereign CDS levels. Hence, Bunds richened versus EURIBOR at the same time as German CDS levels increased.

An analyst who fails to consider these interconnected valuation relations may find the combination of richening Bunds and increasing German CDS opaque and puzzling. But a well-equipped relative value analyst can disentangle these valuation relations explicitly to identify the factors that are driving valuations in these markets. And armed with this knowledge, the analyst can apply these insights to other instruments, potentially uncovering additional relative value opportunities.

THE APPLICATIONS OF RELATIVE VALUE ANALYSIS

Relative value analysis has a number of applications.

Trading

One of the most important applications of relative value analysis is relative value trading, in which various securities are bought and others sold with the goal of enhancing the risk-adjusted expected return of a trading book.

Identifying relatively rich and relatively cheap securities is an important skill for a relative value trader, but additional skills are required to be successful as a relative value trader. For example, rich securities can and often do become richer, while cheap securities can and often do become cheaper. A successful relative value trader needs to be able to identify some of the reasons that securities are rich or cheap in order to form realistic expectations about the likelihood of future richening or cheapening. We discuss this and other important skills throughout this book.

Hedging and Immunization

Relative value analysis is also an important consideration when hedging or otherwise immunizing positions against various risks. For example, consider a flow trader who is sold a position in ten-year (10Y) French government bonds by a customer. This trader faces a number of alternatives for hedging this risk.

He could try to sell the French bond to another client or to an interdealer broker. He could sell another French bond with a similar maturity. He could sell Bund futures contracts or German Bunds with similar maturities. He could pay fixed in an interest rate swap. He could buy payer swaptions or sell receiver swaptions with various strikes. He could sell liquid supranational or agency bonds issued by entities such as the European Investment Bank. Depending on his expectations, he might even sell bonds denominated in other currencies, such as US Treasuries or UK Gilts. Or he might choose to implement a combination of these hedging strategies.

In devising a hedging strategy, a skilled trader will consider the relative valuations of the various securities that can be used as hedging instruments. If he expects Bunds to cheapen relative to the alternatives, he may choose to sell German Bunds as a hedge. And if he believes Bund futures are likely to cheapen relative to cash Bunds, he may choose to implement this hedge via futures contracts rather than in the cash market.

By considering the relative value implications of these hedging alternatives, a skilled flow trader can enhance the risk-adjusted expected return of his book. In this way, the value of the book reflects not only the franchise value of the customer flow but also the relative value opportunities in the market and the analytical skills of the trader managing the book.

Given the increasing competitiveness of running a fixed income flow business, firms that incorporate relative value analysis as part of their business can expect to increase their marginal revenues, allowing them to generate higher profits and/or to offer liquidity to customers at more competitive rates.

Security Selection

In many respects, a long-only investment manager faces many of the same issues as the flow trader in the previous example. Just as a flow trader can expect to enhance the risk-adjusted performance of his book by incorporating relative value analysis into his hedging choices, a long-only investment manager can expect to enhance the risk-adjusted performance of his portfolio by incorporating relative value analysis into his security selection process.

For example, an investment manager who wants to increase his exposure to the 10Y sector of the EUR debt market could buy government bonds issued

by France, Germany, Italy, Spain, the Netherlands, or any of the other EMU member states. Or he could buy Bund futures or receive fixed in a EURIBOR or ESTR interest rate swap. Or he might buy a US Treasury in conjunction with a cross-currency basis swap, thereby synthetically creating a US government bond denominated in euros.

An investment manager who incorporates relative value analysis as part of his investment process is likely to increase his alpha and therefore over time to outperform an otherwise similar manager with the same beta who doesn't incorporate relative value analysis.

THE CRAFT OF RELATIVE VALUE ANALYSIS

Relative value analysis is neither a science nor an art. Rather, it's a craft, with elements of both science and art. For a practitioner to complete the journey from apprentice to master craftsman, he needs to learn to use the tools of the trade, and in this book we introduce these tools along with their foundations in the mathematical science of statistics and in the social science of financial economics.

We also do our best to explain the practical benefits and potential pitfalls of applying these tools in practice. In the development of an apprentice, there is no substitute for repeated use of the tools of the trade in the presence of a master craftsman. So we make every effort in this book to convey the benefit of our experience over many years of applying these tools.

Since financial and statistical models are the tools of the trade for a relative value analyst, it's important that the analyst chooses these tools carefully, with an eye toward usefulness, analytical scope, and parsimony.

Usefulness

In our view, models are neither right nor wrong. Pure mathematicians may be impressed by truth and beauty, but the craftsman is concerned with usefulness. To us, various models have varying degrees of usefulness, depending on the context in which they're applied.

As Milton Friedman reminds us in his 1966 essay "The Methodology of Positive Economics," models are appropriately judged by their implications. The usefulness of a particular model is not a function of the realism of its assumptions but rather of the quality of its predictions.

For relative value analysts, models are useful if they allow us to identify relative misvaluations between and among securities, and if they improve the quality of the predictions we make about the future richening and cheapening of these securities.